

Citation Verification for “Surrogate Inverse-Variance Weighting for Intervention Harvesting on Heavy-Tailed Click Logs”

(internal verification document, not submitted)

This document quotes verbatim from each cited paper the passage that supports the corresponding claim in the CIKM short paper. Goal: let the authors verify that each citation accurately backs the claim it is attached to. Local PDFs of the cited papers live in `paper/cikm/citations/`.

Summary of issues found during verification

Two PDF mismatches need attention:

1. `swaminathan2015snips`: our local file `swaminathan_joachims.2015_nips.pdf` is actually the *ICML 2015* paper “Counterfactual Risk Minimization” by the same authors, not the NeurIPS 2015 SNIPS paper that the bibitem cites. Action: either download the NeurIPS paper from <https://papers.nips.cc/paper/2015/hash/39027dfad5138c9ca0c474d71db915c3-Abstract.html>, or change the bibitem to cite the ICML CRM paper and adjust in-text wording.
2. `yue2010beyond`: our local file `yue.2010_www.pdf` is actually a *different* Yue 2010 paper (SIGIR 2010 “Learning More Powerful Test Statistics for Click-Based Retrieval Evaluation” by Yue, Gao, Chapelle, Zhang, Joachims), not the WWW 2010 paper by Yue, Patel, Roehrig that the bibitem cites. Action: download the correct WWW 2010 paper, or replace the cite (Chuklin 2015 §3.3 / Craswell 2008 cascade discussion already cover related PBM-violation material).

Partial verifications still to complete (load-bearing parts of the cited papers not yet read in detail):

- `bottou2013counterfactual`: §4 variance-decomposition for self-normalised IPS.
- `wang2018position`: experimental Section 6, whether the “PBM doesn’t fully hold on Gmail/Drive” claim is supported by their data.
- `zou2022large`: dataset-construction section, whether the 94% $N = 1$ and 78% ratio statistic is documented there or only computable by us from the released aggregation.
- `craswell2008experimental`: the cascade-model definition matches our usage as a PBM-violation alternative.

Verified (author, title, and load-bearing claim confirmed): `agarwal2019estimating`, `borenstein2009meta`, `chuklin2015clickmodels`, `hesterberg1995`, `joachims2017unbiased`, `kong1992ess`, `owen2013mc`.

Author/title only verified (book or partial PDF, content not in local copy): vdv1998asymptotic (frontmatter only).

No PDF expected (dataset citation): kddcup2012.

1. agarwal2019estimating

Paper claim: The intervention-harvesting estimator of [?] uses uniform per-cell weighting (each cell’s click rate enters the sum with weight 1).

Source location: Section 4.1 “Local Estimators”, PivotOne and AdjacentChain definitions on p. 4–5 of the WSDM 2019 paper.

*Since the click counts $\hat{c}_k^{k,k'}$ and $\hat{c}_{k'}^{k,k'}$ can be treated just like data from an explicit intervention, the same estimators apply. In particular, we observe the interventional set $S_{1,k}$ and can thus use the same estimator that we previously used for the explicit $\text{Swap}(1,k)$ experiment. We call this the **PivotOne** estimator*

$$\hat{p}_k / \hat{p}_1 = \hat{c}_k^{1,k} / \hat{c}_1^{1,k}.$$

*We can similarly adapt the estimator used in [29], which uses a chain of swaps between adjacent positions in the ranking. We call this the **AdjacentChain** estimator $\hat{p}_k / \hat{p}_1 = (\hat{c}_2^{1,2} / \hat{c}_1^{1,2}) \cdot (\hat{c}_3^{2,3} / \hat{c}_2^{2,3}) \cdots (\hat{c}_k^{k-1,k} / \hat{c}_{k-1}^{k-1,k})$.*

Each $\hat{c}$ is defined earlier (Section 3.4) as

$$\hat{c}_k^{k,k'} := \sum_{i=1}^m \sum_{j=1}^{n_i} \sum_{d \in \Omega(q_i^j)} \mathbf{1}[(q_i^j, d) \in S_{k,k'}] \mathbf{1}[rk(d|y_i^j) = k] \frac{c_i^j(d)}{w(q_i^j, d, k)}.$$

The weight $w(q, d, k)$ normalises by ranker-assignment count (propensity-style), and at the cell aggregation level the per-cell click rate C_k^i / N_k^i enters with coefficient 1 (no heteroscedastic re-weighting). **No alternative per-cell weighting (min, harmonic, ...) is discussed in the paper.** Verified.

2. borenstein2009meta

Paper claim: (i) Equal-weighting heteroscedastic estimators is statistically suboptimal by classical inverse-variance theory; (ii) inverse-variance weighting is canonical in meta-analysis; (iii) the count-only IVW optimum restricted to count-only weights is the harmonic mean (a classical result that Borenstein presents in standard form).

Source location: Borenstein, Hedges, Higgins, Rothstein, *Introduction to Meta-Analysis*, Wiley 2009. Sample chapter PDF in `citations/borenstein_2009_meta_chapter.pdf`.

*Borenstein et al. Ch. 11 summary (p. 67): “Under the fixed-effect model all studies in the analysis share a common true effect. The summary effect is our estimate of this common effect size, and the null hypothesis is that this common effect is zero (for a difference) or one (for a ratio). All observed dispersion reflects sampling error, and **study weights are assigned with the goal of minimizing this within-study error.**”*

Ch. 12 (p. 72), under “Performing a random-effects meta-analysis” (the same logic applies to fixed-effect meta-analysis with within-study variance only): “In order to obtain the most precise estimate of the overall mean (to minimize the variance) we compute a weighted mean, where the weight assigned to each study is the inverse of that study’s variance.”

Ch. 13 (p. 78, “Estimating the Summary Effect”): “Under the fixed-effect model we assume that the true effect size for all studies is identical, and the only reason the effect size varies between studies is sampling error (error in estimating the effect size). Therefore, when assigning weights to the different studies we can largely ignore the information in the smaller studies since we have better information about the same effect size in the larger studies.”

Verified. The cite supports both “equal-weighting heteroscedastic estimators is statistically suboptimal” and “inverse-variance weighting is canonical in meta-analysis”. (Aitken 1934 is the original GLS / BLUE source; Borenstein covers the same content in modern textbook form.)

3. bottou2013counterfactual

Paper claim: The closest off-policy analogue to our plug-in $\tilde{V}$ framework is SNIPS-style variance analysis of self-normalised IPS in counterfactual evaluation; weight heterogeneity (not sample count) governs effective precision on logged feedback.

Source location: Bottou, Peters, Quiñero-Candela, Charles, Chickering, Portugaly, Ray, Simard, Snelson, “Counterfactual Reasoning and Learning Systems: The Example of Computational Advertising”, JMLR 14:3207–3260, 2013. PDF: [citations/bottou_2013_jmlr.pdf](#). Author/title verified from p. 1.

From the abstract: “This work shows how to leverage causal inference to understand the behavior of complex learning systems interacting with their environment and predict the consequences of changes to the system. [...] illustrated by experiments on the ad placement system associated with the Bing search engine.”

Author/title verified. **TODO: verify the specific variance-decomposition / weight-heterogeneity claim in their §4 (Counterfactual estimation).** The paper is 50 pages; the most-cited result is the importance-sampling variance bound which is consistent with our usage as “off-policy analogue.”

4. chuklin2015clickmodels

Paper claim: Click-model survey including PBM and other click models; canonical reference for click-model background.

Source location: Chuklin, Markov, de Rijke, *Click Models for Web Search*, Morgan & Claypool 2015 (author’s version PDF in [citations/chuklin_2015_clickmodels.pdf](#)). Table of contents confirms PBM and Cascade among the basic click models.

From the Contents (p. ii): “Chapter 3 Basic Click Models: 3.3 Position-based Model (PBM), 3.4 Cascade Model (CM), 3.5 User Browsing Model (UBM), 3.6 Dependent Click Model (DCM), 3.7 Click Chain Model (CCM), 3.8 Dynamic Bayesian Network Model (DBN).”

Verified. The book is the canonical click-model survey and explicitly covers PBM and its alternatives. Use to back the claim “click models beyond PBM”. **TODO** if used to back the explicit “PBM violations documented in real logs” claim: a specific quote from their experimental chapters comparing models on real data would be needed.

5. craswell2008experimental

Paper claim: Click position-bias models; the cascade model and basic position bias hypotheses.

Source location: Craswell, Zoeter, Taylor, Ramsey, WSDM 2008. PDF: [citations/craswell_2008_wsdm.pdf](#).

From the abstract: “Search engine click logs provide an invaluable source of relevance information, but this information is biased. A key source of bias is presentation order: the probability of click is influenced by a document’s position in the results page. This paper focuses on explaining that bias, modelling how probability of click depends on position. We propose four simple hypotheses about how position bias might arise. We carry out a large data-gathering effort, where we perturb the ranking of a major search engine, to see how clicks are affected. We then explore which of the four hypotheses best explains the real-world position effects, and compare these to a simple logistic regression model. The data are not well explained by simple position models, where some users click indiscriminately on rank 1 or there is a simple decay of attention over ranks. A ‘cascade’ model, where users view results from top to bottom and leave as soon as they see a worthwhile document, is our best explanation for position bias in early ranks.”

The paper does *not* document the PBM as such (PBM is later formalisation by Chuklin/Wang). We use this cite for “three explicit PBM violation models – trust bias, cascade, and position-dependent relevance”. The cascade model is introduced here; PBM and trust bias are formalised in Chuklin 2015 / Joachims 2017. **TODO: confirm cascade definition matches our usage in §5.**

6. hesterberg1995

Paper claim: Self-normalised weighted ratio estimators are standard in importance sampling; heterogeneous-weight importance sampling variance analysis.

Source location: Tim Hesterberg, “Weighted Average Importance Sampling and Defensive Mixture Distributions”, *Technometrics* 37(2):185–194, 1995. PDF: [citations/hesterberg_1995_technometrics.pdf](#)

*From the abstract: “Importance sampling uses observations from one distribution to estimate for another distribution by weighting the observations. Including the target distribution as one component of a mixture distribution bounds the weights and makes importance sampling more reliable. **The usual importance-sampling estimate is a weighted average with weights that do not sum to 1.** We discuss simple normalization and other, more efficient normalization methods.”*

§1 Introduction (p. 185): “The integration estimate is a weighted average with weights that do not sum to 1. Let $W(x) = f(x)/g(x)$ be the weight function, based on the likelihood ratio between the target and design distributions; then the empirical distribution function corresponding to the integration estimate has weight $V_{int,i} = n^{-1}W_i$ on the output $Q(X_i)$ from the i th replication, where $W_i = W(X_i)$. The sum of the empirical weights is $\bar{W}$, which has expected value 1 but nonzero variance.”

Verified. The paper is exactly about the variance of self-normalised weighted IS estimators, which is what we cite for.

7. joachims2017unbiased

Paper claim: Position bias systematically corrupts click signals used for learning-to-rank; IPS is the standard counterfactual estimator.

Source location: Joachims, Swaminathan, Schnabel, “Unbiased Learning-to-Rank with Biased Feedback”, WSDM 2017. PDF: [citations/joachims_2017_wsdm.pdf](#).

From the abstract: “While implicit feedback has many advantages (e.g., it is inexpensive to collect, user centric, and timely), **its inherent biases are a key obstacle to its effective use. For example, position bias in search rankings strongly influences how many clicks a result receives, so that directly using click data as a training signal in Learning-to-Rank (LTR) methods yields sub-optimal results.** To overcome this bias problem, we present a counterfactual inference framework that provides the theoretical basis for unbiased LTR via Empirical Risk Minimization despite biased data.”

§2 Related Work: “Our approach uses inverse propensity scoring (IPS), originally employed in survey sampling [7] and causal inference from observational studies [19], but more recently also in whole page optimization [29], IR evaluation with manual judgments [13, 21], and recommender evaluation [13, 12].”

Verified. Cite supports both “position bias systematically corrupts click signals” (intro) and “IPS / counterfactual framework for ULTR” (the standard reference).

8. kddcup2012 (dataset, no PDF)

Paper claim: Dataset reference for the KDD Cup 2012 Track 2 click log.

Source location: Tencent / KDD Cup 2012 competition release. <https://www.kaggle.com/c/kddcup2012-track2>

(Dataset citation; no paper. The dataset statistics in the paper were computed by the authors directly from the released training file.)

9. kong1992ess

Paper claim: Effective sample size $\text{ESS}(\omega) = (\sum_i \omega_i)^2 / \sum_i \omega_i^2$.

Source location: Kong, “A note on importance sampling using standardized weights”, Tech. Rep. 348, U. Chicago Statistics, 1992. PDF: [citations/kong_1992_tr348.pdf](#). Also re-derived in Owen 2013 §9.3 Eq. (9.13).

*From Owen Ch. 9, Eq. (9.13) (which cites Kong 1992): “Setting $\text{Var}(S_w) = \sigma^2/n_e$ and solving for n_e yields the **effective sample size** $n_e = (\sum_{i=1}^n w_i)^2 / \sum_{i=1}^n w_i^2 = n\bar{w}^2/\bar{w}^2$.”*

Verified via Owen §9.3.

10. owen2013mc

Paper claim: (i) The textbook closed-form asymptotic variance of the self-normalised weighted ratio estimator $\hat{V}(\omega)$; (ii) Effective sample size.

Source location: Owen, *Monte Carlo Theory, Methods and Examples*, Ch. 9 (“Importance sampling”), §9.2 “Self-normalized importance sampling” (pp. 8–10) and §9.3 “Importance sampling diagnostics” (pp. 11–12). PDF: [citations/owen_mc_ch09_importance.pdf](#).

*Owen §9.2, Eq. (9.6)–(9.9): “Consider the **self-normalized importance sampling estimate** $\tilde{\mu}_q = \sum_i f(X_i)w_u(X_i) / \sum_i w_u(X_i)$. To place a confidence interval around $\tilde{\mu}_q$ we use the delta method and equation (2.30) from §2.7 for the variance of a ratio of means. Applying that formula to (9.7) yields an approximate variance for $\tilde{\mu}_q$ of $\widehat{\text{Var}}(\tilde{\mu}_q) = (1/n) \mathbb{E}_q((f(X)w(X) - \mu w(X))^2) / \mathbb{E}_q(w(X))^2 = \sigma_{q,sn}^2/n$. For a computable variance estimate, we employ w_u instead of w , getting $\widehat{\text{Var}}(\tilde{\mu}_q) = \sum_i w_i^2 (f(x_i) - \tilde{\mu}_q)^2$.”*

*Owen §9.3, Eq. (9.13): “Setting $\text{Var}(S_w) = \sigma^2/n_e$ and solving for n_e yields the **effective sample size** $n_e = (\sum_{i=1}^n w_i)^2 / \sum_{i=1}^n w_i^2$.”*

Verified.

11. swaminathan2015snips

Paper claim: SNIPS is the off-policy analogue of our plug-in $\tilde{V}$ framework: same self-normalised weighted ratio variance applied to counterfactual evaluation.

Source location: Swaminathan & Joachims, NeurIPS 2015, *The Self-Normalized Estimator for Counterfactual Learning*.

MISMATCH: our local file [swaminathan-joachims-2015_nips.pdf](#) is actually the ICML 2015 paper “Counterfactual Risk Minimization: Learning from Logged Bandit Feedback” by the same authors, not the NeurIPS SNIPS paper. The two are related (CRM uses variance-regularised IPS; SNIPS introduces the self-normalised IPS estimator) but the SNIPS-specific construction is in the NeurIPS paper, which we do not have locally. **Need to download the NeurIPS version to verify the SNIPS claim.**

*From the ICML CRM paper we have (§3): “Third, importance sampling typically estimates $\hat{R}^M(h)$ of different hypotheses $h \in \mathcal{H}$ with **vastly different variances**. Consider two hypotheses h_1 and h_2 , where h_1 is similar to h_0 , but where h_2 samples predictions that were not well explored by h_0 . Importance sampling gives us low-variance estimates for $\hat{R}^M(h_1)$, but highly variable estimates for [...]”*

This supports the loose claim that “weight heterogeneity, not sample count, governs effective precision” but not specifically SNIPS. Fix: either download the NIPS 2015 paper (arXiv preprint <https://arxiv.org/abs/1502.02362> is the ICML one; the NIPS paper is at <https://papers.nips.cc/paper/2015/hash/39027dfad5138c9ca0c474d71db915c3-Abstract.html>), or change the bibitem to cite the ICML CRM paper instead and adjust the in-text wording.

12. vdv1998asymptotic

Paper claim: Delta method.

Source location: van der Vaart, *Asymptotic Statistics*, Cambridge 1998, Ch. 3 “Delta method”. Frontmatter PDF only: [citations/vdv_1998_frontmatter.pdf](#). Ch. 3 content not in our local copy.

(Citation is the standard reference for the delta method; we verify only author/title/publisher from the frontmatter. Wasserman 36-705 lec 3 [citations/wasserman_36705_lec03.pdf](#) provides a freely available equivalent.)

13. wang2018position

Paper claim: (i) PBM EM joint estimation of relevance and propensities; (ii) PBM violations on real personal-search logs.

Source location: Wang, Golbandi, Bendersky, Metzler, Najork, “Position Bias Estimation for Unbiased Learning to Rank in Personal Search”, WSDM 2018. PDF: [citations/wang_2018_wsdm.pdf](#).

*From the abstract: “We propose a **regression-based Expectation-Maximization (EM) algorithm** that is based on a position bias click model and that can handle highly sparse clicks in personal search. We evaluate our EM algorithm and the extracted bias in the learning-to-rank setting.”*

§2 Related Work: “Recently, Wang et al. [31] and Joachims et al. [20] introduced the unbiased learning-to-rank framework by treating the bias as a counterfactual effect. Inside the framework, the critical component is to quantify the click bias, rather than estimate the $\langle \text{query}, \text{document} \rangle$ relevance in click models.”

(i) Verified: EM estimation of position bias is the core contribution.

(ii) Partial: the abstract and intro do not explicitly document Gmail/Drive PBM violations as a finding. The claim “PBM doesn’t fully hold on Gmail and Drive search logs” should be located in their experimental section before we use it as direct evidence. Currently in the paper we cite Wang 2018 alongside other PBM violation work; if Section 6 of Wang 2018 does not contain explicit

PBM-violation evidence, we should either restrict the cite to the EM claim or substitute Chuklin 2015 / Yue 2010 for the PBM-violation claim.

14. yue2010beyond

Paper claim: Documents that real click data has effects beyond position (attractiveness, trust bias, etc.) – i.e. PBM-violating structure.

Source location: Yue, Patel, Roehrig, “Beyond Position Bias: Examining Result Attractiveness as a Source of Presentation Bias in Clickthrough Data”, WWW 2010.

MISMATCH: our local file `yue_2010_www.pdf` is actually a *different* Yue 2010 paper: “Learning More Powerful Test Statistics for Click-Based Retrieval Evaluation” by Yue, Gao, Chapelle, Zhang, Joachims (SIGIR 2010, pp. 507–). The SIGIR paper is about interleaving test statistics, not about attractiveness/PBM violations. **Need to download the actual WWW 2010 paper** by Yue, Patel, Roehrig.

TO BE VERIFIED on the correct PDF (WWW 2010 Yue, Patel, Roehrig). The standard summary of that paper is: result attractiveness (title, snippet quality) creates click-rate variation that is not explained by position alone, demonstrating that PBM is an approximation and that result-level features influence clicks beyond position.

15. zou2022large

Paper claim: Baidu-ULTR is a large search dataset for unbiased learning-to-rank; the released aggregation has 94% of cells with $N = 1$ and 78% of adjacent-pair ratios in $[0.5, 2]$, which we attribute to the per-cell deduplication in the public release.

Source location: Zou et al., NeurIPS 2022. PDF: `citations/zou.2022_neurips.pdf`.

TO BE FILLED IN. (Dataset construction and deduplication details should be in their data-preparation section; our 94%/78% numbers were computed by us directly on the released aggregation.)